



Fig. 2: BPL electromagnetic model

it to the wireless access point (AP) located on the pole (Fig. 1).

For end-to-end BPL, bypass boxes and LV couplers must be installed on all LV lines, and in-house BPL modems are required. For hybrid BPL, bypass boxes with wireless conversion boards, wireless APs and existing standard wireless user modems are required.

Architecture and standards

Based on seven-layer open system interconnection model defined in ISO-7498, BPL systems are designed to work on a two-layer architecture comprising physical (PHY) and medium access control (MAC) layers.

The PHY layer defines electrical and physical specifications for devices, i.e., relationship between a device and a physical medium. It includes all the electrical power line distribution system and the in-home electrical wiring down to the wall sockets. The major functions of PHY layer include establishment and termination of a connection to a communication medium, communication flow control, modulation and coding.

MAC layer, consisting of MAC lower sub-layer and logical link control layer, provides an interface between PHY layer and higher layers for connection to physical media.

At initial stage, manufacturers of BPL systems have developed their own proprietary solutions for injectors, repeaters, extractors and couplers. These solutions have been implemented in a variety of system architectures, which feature different modulation techniques and designs at PHY, MAC and LLC layers. Such systems have raised

incompatibility issues.

In order to combat channel impairments (noise, multipath, strong channel selectivity, non-linear channel characteristics), a number of different technologies at PHY layer have been employed that range from spread spectrum to orthogonal frequency-division multiplexing (OFDM).

OFDM is a type of frequency-division multiplexing system that provides better channel throughput, better spectrum efficiency and robustness against frequency selectivity because all of the underlying sub-carriers are orthogonal to one another.

There are numerous modulation and access design techniques that can be used in association with OFDM. These include binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), 16-quadrature amplitude modulation (QAM) and 6-bit, 64-constellation QAM. Various forward error correction codes are also used.

Since BPL access networks operate in a shared transmission medium where subscribers compete to use the same transmission resources, MAC has been designed for point-to-multipoint applications and is based on collision-sense multiple-access with collision avoidance (CSMA/CA). MAC layer also describes how secure communications are delivered, by using secure key exchange during authentication and encryption (using advanced encryption standard or data encryption standard) during data transfer.

In order to streamline the functionality of BPL systems, Open Power line communication European Research Alliance, European Telecommunications Standards Institute, Universal Powerline Association, European Committee for Electrotechnical Standardization, HomePlug Powerline Alliance and Institute of Electrical and Electronics Engineers (IEEE) have developed standards for BPL systems.

The IEEE has constituted IEEE P1675 'Standard for Broadband over Power

Line Hardware', IEEE P1775 'Powerline Communication Equipment—Electromagnetic Compatibility (EMC) Requirements—Testing and Measurement Methods' and IEEE P1901 'Draft Standard for Broadband over Power Line Networks: Medium Access Control and Physical Layer Specifications.'

HomePlug Powerline Alliance has developed in-house BPL specifications: HomePlug 1.0 for speeds up to 14 Mbps and HomePlug AV for speeds greater than 100 Mbps.

BPL issues

Interference issues. Most BPL systems are designed to operate in the frequency spectrum from 2 to 30 MHz, but occasionally up to 80 MHz, using MV and LV power distribution network lines. The frequency spectrum from 2 to 30 MHz constitutes a limited natural resource that includes the HF band (3 to 30 MHz), which is being used for many decades by shortwave radio stations, military, aviation agencies, etc. Above the HF band, the frequency spectrum from 30 to 54 MHz is reserved for use by public service and business communications. The spectrum from 54 to 80 MHz hosts television channels (channel numbers 2 to 5) with a small segment used for some other emergency services.

However, because BPL uses some of the radio frequencies used for over-the-air radio systems, mutual interference is a major problem. In order to mitigate interference at some specific frequency, OFDM may be used as it has the ability to notch these specific frequencies.

Moreover, power lines are unshielded and act as radiating antennae for the signals they carry, resulting in interference with communication systems of broadcast radio, military, aviation agencies, etc.

Electromagnetic fields associated with access-BPL system are shown in Fig. 2 and can be categorised as guided mode, coupler fields and radiation mode. Guided mode serves to transport signal energy along the line. Signal energy decays rapidly in